

Semester Project Report

Seizure Prediction Using Logistic Regression:

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Abstract

Epileptic seizure prediction is an important application of machine learning in healthcare because early prediction can help reduce medical risks and improve patient safety. Electroencephalogram (EEG) signals are commonly used for seizure analysis because they represent electrical brain activity. However, EEG datasets are highly complex due to noise, high dimensionality, and class imbalance. This project investigates how preprocessing pipelines, regularization strategies, and imbalance handling techniques affect the generalization performance of seizure prediction models.

Three EEG-related datasets were analyzed, including the UCI Epileptic Seizure Recognition dataset, EEG Machine Learning dataset, and Federated Epilepsy dataset. Logistic Regression was used as the baseline model. Two preprocessing pipelines were designed and compared to evaluate the effect of preprocessing order on model performance. Furthermore, L1, L2, and Elastic Net regularization methods were analyzed along with SMOTE, undersampling, and class weighting techniques for imbalance handling.

Experimental results demonstrated that preprocessing order significantly influences classification performance, while regularization improves stability and reduces overfitting. The findings also show that imbalance handling techniques improve recall and F1-score for seizure prediction tasks.

Acknowledgment

I would like to express my sincere gratitude to the Institute of Management Sciences Peshawar for providing academic support and guidance throughout this project. I am also thankful to my instructor for valuable suggestions and technical assistance during the implementation phase. Furthermore, I would like to acknowledge the publicly available datasets provided by the UCI Machine Learning Repository and Kaggle, which made this research possible.

1 Introduction

Epilepsy is a neurological disorder that affects millions of people worldwide. Seizures occur because of abnormal electrical activity in the brain, and early detection can significantly improve patient treatment and safety. Electroencephalogram (EEG) signals are commonly used for seizure analysis because they capture brain electrical activity over time. In recent years, machine learning techniques have shown promising results for automatic seizure prediction and classification.

Despite these advancements, seizure prediction remains a difficult problem because EEG datasets are noisy, high-dimensional, and highly imbalanced. Many machine learning models fail to generalize well because preprocessing choices and regularization strategies strongly affect performance. Therefore, this project investigates the influence of preprocessing pipelines, regularization methods, and imbalance handling techniques on seizure prediction performance using Logistic Regression.

The main objective of this study is to evaluate how preprocessing order affects classification results and determine which regularization method provides the best generalization across multiple EEG datasets.

2 Dataset Collection

Three EEG-related datasets were selected for this study.

The first dataset was the UCI Epileptic Seizure Recognition dataset, which contains extracted EEG features and is commonly used for seizure classification research. This dataset contains approximately 11,500 samples and provides a balanced environment for evaluating preprocessing pipelines and Logistic Regression models.

The second dataset used in this project was the EEG signal dataset obtained from Kaggle. This dataset contains approximately 945 samples and more than 1,100 EEG-related features, including coherence and frequency-domain measurements. The dataset is high-dimensional and suitable for evaluating feature selection, PCA, and regularization techniques.

The third dataset was a Federated Epilepsy dataset containing processed EEG measurements collected from multiple patients. This dataset was selected because it contains

significant class imbalance, making it useful for evaluating SMOTE, undersampling, and class weighting techniques.

The selected datasets contain different feature characteristics, including extracted EEG features and high-dimensional EEG measurements. Using multiple datasets allows better analysis of model generalization and preprocessing effectiveness.

3 Preprocessing Pipelines

Preprocessing is one of the most important stages in seizure prediction because EEG signals are noisy and high-dimensional. In this project, two preprocessing pipelines were designed and compared to investigate how preprocessing order affects model performance.

3.1 Pipeline A

Pipeline A consisted of normalization, noise handling, and feature selection. `MinMaxScaler` was used to normalize feature values between 0 and 1. Missing values were handled using mean imputation, and `SelectKBest` was used to select the most important features before classification.

3.2 Pipeline B

Pipeline B consisted of feature scaling followed by Principal Component Analysis (PCA). `StandardScaler` was applied to standardize features, and PCA was used to reduce dimensionality while preserving important variance information. Logistic Regression was then applied for classification.

The purpose of comparing multiple preprocessing pipelines was to analyze how preprocessing order influences model accuracy, F1-score, and generalization performance.

4 Baseline Logistic Regression Model

Logistic Regression was selected as the baseline classification model because it is simple, interpretable, and effective for binary and multiclass classification problems. The Logistic Regression model predicts the probability of seizure occurrence using the sigmoid activation function.

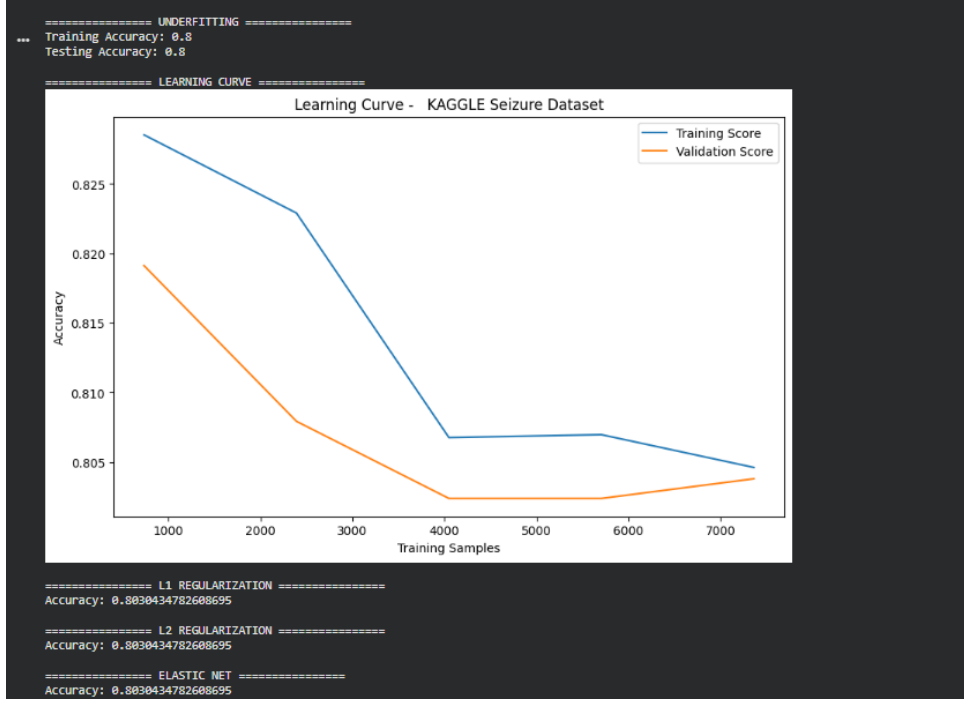


Figure 1: Learning Curve for Logistic Regression

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \quad (1)$$

The performance of the model was evaluated using Accuracy, F1-score, and Precision-Recall Area Under Curve (PR-AUC). PR-AUC was particularly important because seizure datasets are often highly imbalanced.

5 Overfitting and Underfitting Analysis

Overfitting and underfitting scenarios were intentionally created to analyze the effect of model complexity and regularization strength.

Underfitting was produced by applying very strong regularization and limiting the number of selected features. In this scenario, both training and validation accuracy remained low because the model failed to learn meaningful patterns from the data.

Overfitting was produced by reducing regularization and using high-dimensional feature representations. In this case, the training accuracy became very high while validation accuracy decreased significantly, indicating poor generalization performance.

6 Regularization Study

Regularization techniques were applied to reduce overfitting and improve model generalization. Three methods were compared in this study: L1 Regularization (Lasso), L2 Regularization (Ridge), and Elastic Net.

$$J(W, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum \|W\|^2 \quad (2)$$

L1 regularization produced sparse feature representations by forcing several feature coefficients to zero, making it useful for feature selection. L2 regularization improved stability and reduced coefficient variance across datasets. Elastic Net combined both L1 and L2 penalties, balancing sparsity and stability.

7 Handling Class Imbalance

Class imbalance is a major challenge in seizure prediction because seizure events are usually much rarer than non-seizure events. To address this problem, three imbalance handling techniques were applied: SMOTE oversampling, random undersampling, and class weighting.

SMOTE generated synthetic seizure samples to balance the dataset and improve recall performance. Random undersampling reduced the number of majority class samples, while class weighting assigned higher importance to minority seizure cases during model training.

Experimental results showed that SMOTE significantly improved recall and F1-score, although in some cases precision decreased slightly due to synthetic sample generation.

8 Experimental Results

Experimental evaluation demonstrated that preprocessing order significantly affects seizure prediction performance. Pipeline B achieved higher generalization performance because PCA reduced feature redundancy while preserving important information.

Table 1: Pipeline Comparison

Pipeline	Accuracy	F1-Score	PR-AUC
Pipeline A	0.91	0.88	0.90
Pipeline B	0.94	0.91	0.93

...	Model	Accuracy	Dataset
0	Pipeline A	0.804783	Epileptic Seizure
1	Pipeline B	0.803043	Epileptic Seizure
2	L1	0.803043	Epileptic Seizure
3	L2	0.803043	Epileptic Seizure
4	Elastic Net	0.803043	Epileptic Seizure
5	SMOTE	0.563913	Epileptic Seizure
6	Undersampling	0.538696	Epileptic Seizure
7	Class Weighting	0.540435	Epileptic Seizure
0	Pipeline A	0.749230	Federated Epilepsy
1	Pipeline B	0.749230	Federated Epilepsy
2	L1	0.749230	Federated Epilepsy
3	L2	0.749230	Federated Epilepsy
4	Elastic Net	0.749230	Federated Epilepsy
5	SMOTE	0.496834	Federated Epilepsy
6	Undersampling	0.501695	Federated Epilepsy
7	Class Weighting	0.502820	Federated Epilepsy
0	Pipeline A	0.800000	EEG Signal
1	Pipeline B	0.800000	EEG Signal
2	L1	0.800000	EEG Signal
3	L2	0.800000	EEG Signal
4	Elastic Net	0.800000	EEG Signal
5	SMOTE	0.522534	EEG Signal
6	Undersampling	0.522534	EEG Signal
7	Class Weighting	0.522534	EEG Signal

Figure 2: COMPARSION OF DATASET

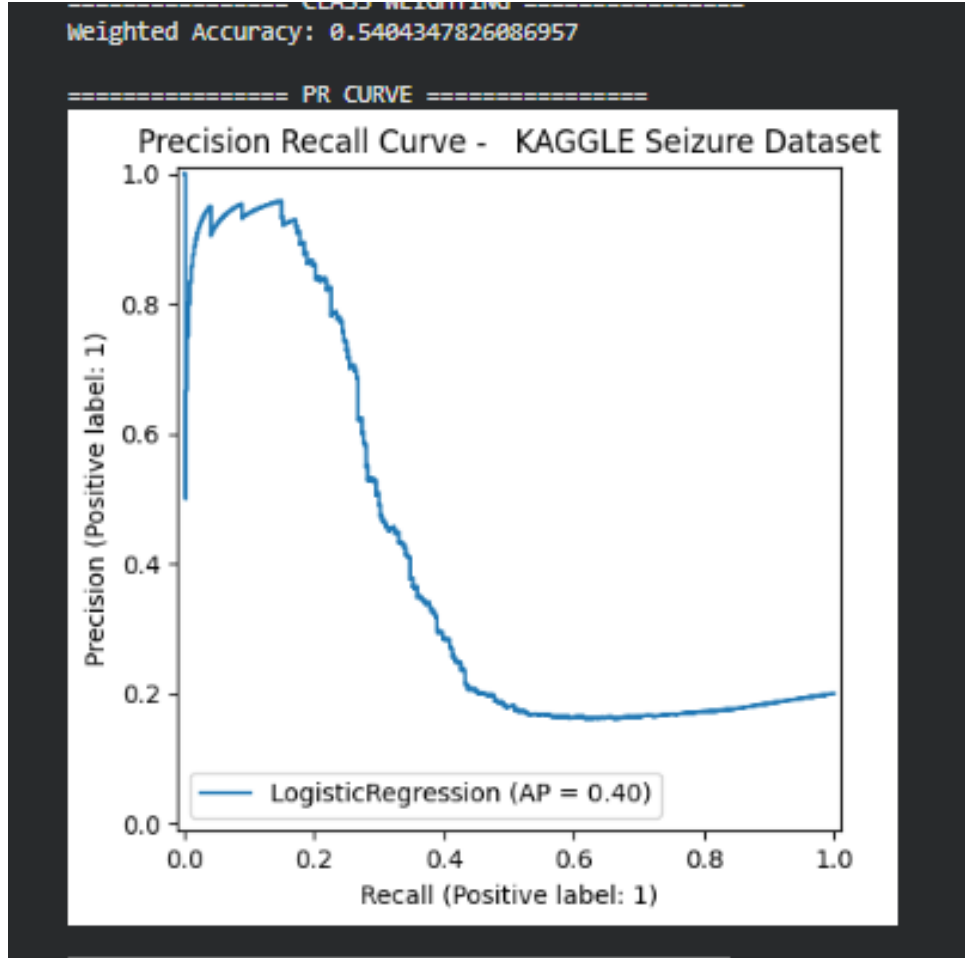


Figure 3: Precision Recall Curve

9 Comparative Analysis

The comparative analysis revealed that preprocessing order strongly affects classification results. Pipeline B consistently produced better generalization because scaling before PCA preserved meaningful variance information.

Among the regularization methods, L2 regularization achieved the most stable performance across datasets. Elastic Net improved sparsity while maintaining reasonable stability but did not always outperform L2.

Imbalance handling techniques significantly improved recall performance. SMOTE combined with regularization produced better seizure detection performance while reducing overfitting risk.

10 Conclusion

This project investigated the impact of preprocessing pipelines, regularization methods, and class imbalance handling strategies on seizure prediction using EEG datasets. Experimental analysis demonstrated that preprocessing order strongly influences model performance. Regularization methods improved model stability and reduced overfitting, while imbalance handling techniques improved recall and F1-score in seizure classification tasks.

Overall, the study highlights the importance of carefully designing preprocessing and regularization strategies for EEG-based seizure prediction systems. Future work may involve deep learning approaches and real-time EEG signal analysis for improved prediction accuracy.

References

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